Formelsammlung ET

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Rechnungen mit spezifischen Widerstand

$R = \frac{\rho \cdot l}{A}; R = \frac{l}{\kappa \cdot A}$ mit $\kappa = \frac{1}{\rho}$

Arbeit und Leistung im Gleichstrom

$W = U \cdot Q; W = U \cdot I \cdot t$

$P = \frac{W}{t}; P = U \cdot I = \frac{U^2}{R} = I^2 \cdot R$

Reihenschaltung:

$R = R_1 + \cdots + R_n; U = U_1 + \cdots U_n; \frac{U_1}{U_2} = \frac{R_1}{R_2}\ ; \frac{U_1}{U} = \frac{R_1}{R_1 + \cdots + R_n} = \frac{R_1}{R}$

Parallelschaltung:

$$I = I_1 + \cdots + I_n;\ R = \frac{R_1 \cdot R_2}{R_1 + R_2};\ \frac{I_1}{I_2} = \frac{R_2}{R_1};\ I_1 = I \cdot \frac{R_2}{R_1 + R_2};\ I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

Dreieck zu Stern:

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

Stern zu Dreieck:

$$R_{12} = R_1 + R_2 + \frac{R_1 \cdot R_2}{R_3}$$

Kondensator:

$Q = C \cdot U$

Wechselstrom:

$$U = \frac{\hat{u}}{\sqrt{2}};\ I = \frac{\hat{\imath}}{\sqrt{2}}$$

komplexe Rechnungen:

$$\underline{Z} = R + j \cdot X$$

$$|\underline{Z}| = Z = \sqrt{R^2 + X^2}$$

$$\varphi = \arg(\arctan) \frac{X}{R}$$

$$\underline{Z} = Z \cdot e^{j \cdot \varphi}$$

$$\underline{Z}_1 \cdot \underline{Z}_2 = Z_1 \cdot Z_2\, e^{j(\varphi_1 + \varphi_2)}$$

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{Z_1}{Z_2}\, e^{j(\varphi_1 - \varphi_2)}$$

Temperatur abhängiger Widerstand:

$R = R_{20} \cdot (1 + a_{20} \cdot \Delta\vartheta)$

<u>Elektronik:</u>

Verstärkung: $I_C = I_B \cdot B$

$v_i = \frac{\Delta i_C}{\Delta i_B}; v_u = \frac{\Delta u_C}{\Delta u_B}$

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$$\tau = R \cdot C = \frac{L}{R}$$

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Transistor:

$I_C = \frac{U_B - U_{CE}}{R_C}; R_V = \frac{U_B - U_q}{I_q + I_B}$

OPV:

$$v = \frac{U_a}{U_e} = -\frac{R_k}{R_1};\ R_k = -v \cdot R_1$$

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{Z_1}{Z_2} e^{j(\varphi_1 - \varphi_2)}$$

Spule:

$$I = \frac{U}{\omega L};\ X_L = \omega L; Reaktanz: B_L = -\frac{1}{\omega L}$$

$$Impedanz:\ jX_L = j\omega L$$
$$Admittanz:\ j \cdot B_L = \frac{1}{jX_L} = -j\frac{1}{X_L} = -j\frac{1}{\omega L}$$

Kondensator:

$$I = \omega \cdot C \cdot U;\ X_C = \frac{1}{\omega C};\ B_C = \omega C$$

$$Admittanz:\ jB_C = j\omega C$$
$$Impedanz:\ jX_C = \frac{1}{jB_C} = \frac{1}{j\omega C} = -j\frac{1}{\omega C}$$

Scheinwiderstand:

$$Z = \frac{U}{I} = \sqrt{R^2 + (\omega L)^2}$$

$$I = \frac{U}{\sqrt{R^2 + (\omega L)^2}}$$

Tiefpass:

$$\frac{U_2}{U_1} = \frac{1}{\sqrt{1 + (R\omega C)^2}}$$

$$f_g = \frac{1}{2\pi RC} = \frac{\mathrm{R}}{2\pi L}\ (auch\ bei\ Hochpass\ so)$$

Hochpass:

$$\frac{U_2}{U_1} = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}}$$

Schwingkreis:

$$f_{res} = \frac{1}{2\pi\sqrt{LC}};\ \omega_r = \frac{1}{\sqrt{LC}};\ X_r = \sqrt{\frac{L}{C}}$$

$$Q = \frac{U_L}{U} = \frac{U_C}{U} = \frac{\omega_r L}{R} = \frac{1}{R\omega_r C};\ d = \frac{1}{Q} = \frac{R}{\omega_r L}$$